



Customer Shopping Behavior Analysis

Uncovering spending patterns, customer segments, and product preferences from **3,900 transactions** to guide strategic business decisions.

Dataset at a Glance

3,900

Purchases

Transactional records across all product categories

18

Columns

Demographics, purchase details & shopping behavior

37

Missing Values

In Review Rating — imputed by category median

\$59.76

Avg Purchase

Average transaction value across the dataset

Key features include customer demographics (Age, Gender, Location, Subscription Status), purchase details (Item, Category, Amount, Season), and shopping behavior (Discount, Promo Code, Frequency, Review Rating, Shipping Type).



Data Preparation in Python

Cleaning & Standardization

Loaded via pandas, explored with `df.info()` and `describe()`. Missing `Review_Rating` values imputed by category median. Columns renamed to `snake_case`.

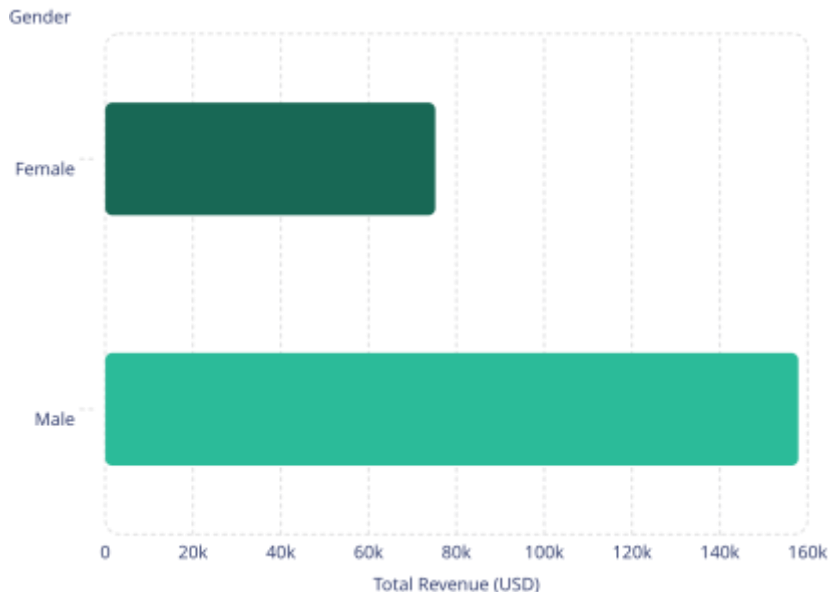
Feature Engineering

Created `age_group` by binning customer ages and `purchase_frequency_days` from purchase data. Dropped redundant `promo_code_used` column.

Database Integration

Cleaned DataFrame loaded into PostgreSQL for structured SQL analysis.

Revenue & Spending Insights



Revenue by Gender

Male customers generated over 2x the revenue of female customers (\$157,890 vs. \$75,191).

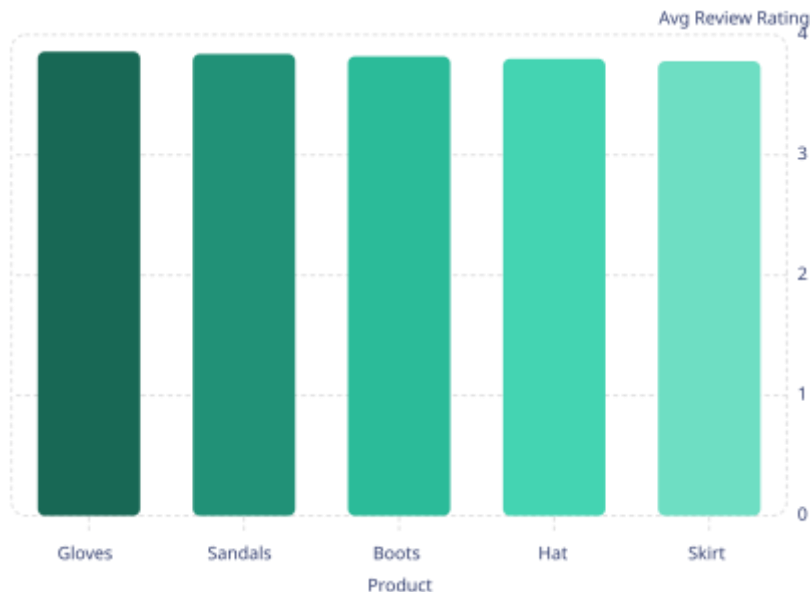
Shipping Type Comparison

Express shipping averaged \$60.48 vs. Standard at \$58.46 — a modest but consistent premium.

High-Spending Discount Users

839 customers used discounts yet still spent above the average purchase amount.

Top Products & Categories



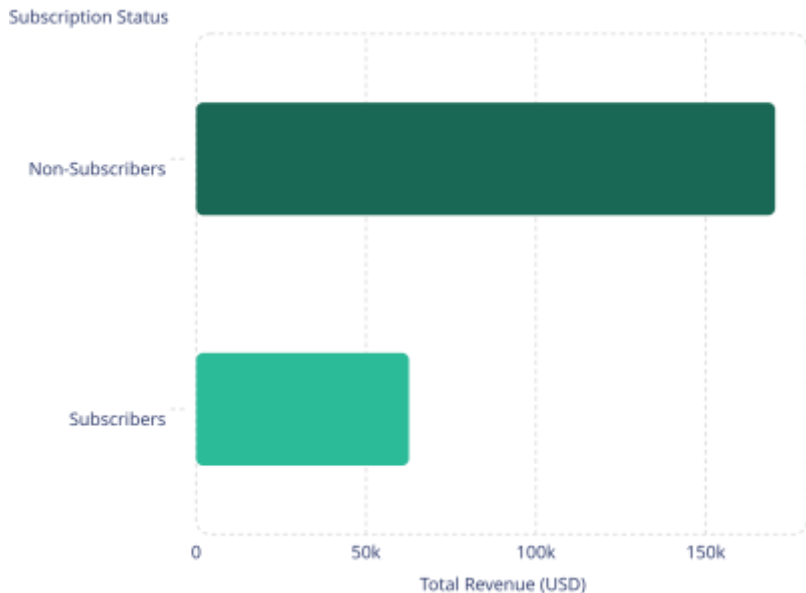
Top 5 by Rating

Gloves lead with a **3.86** average rating. Overall dataset average is **3.75**.

Top Products per Category

- **Accessories:** Jewelry (171), Sunglasses (161), Belt (161)
- **Clothing:** Blouse (171), Pants (171), Shirt (169)
- **Footwear:** Sandals (160), Shoes (150), Sneakers (145)
- **Outerwear:** Jacket (163), Coat (161)

Subscription Behavior



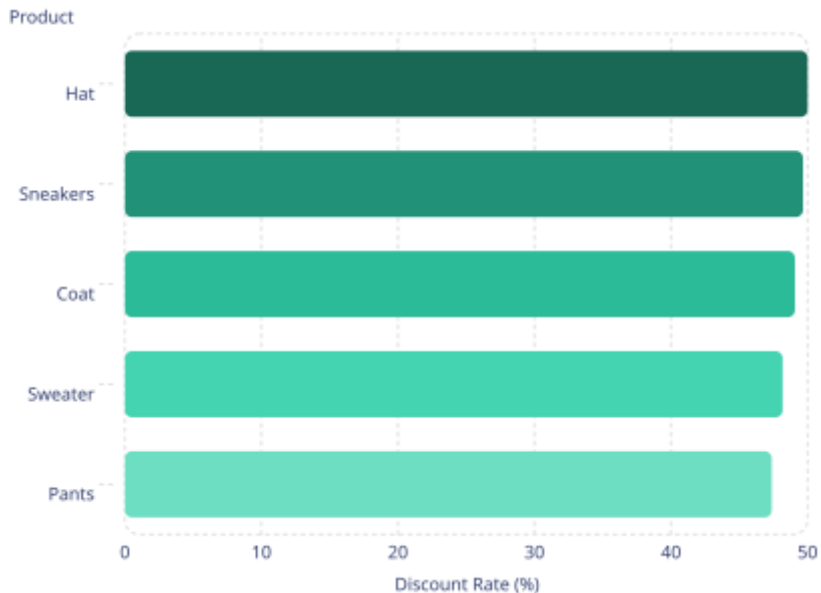
Subscribers vs. Non-Subscribers

Non-subscribers drive **73% of revenue** (\$170,436) despite nearly identical average spend (~\$59.50). Subscribers represent only **27%** of customers (1,053 of 3,900).

Repeat Buyers & Subscriptions

Among customers with **5+ purchases**, 958 are subscribers vs. 2,518 non-subscribers — suggesting subscription adoption lags among repeat buyers.

Discount-Dependent Products



Products Most Reliant on Discounts

The top 5 discount-dependent products all show ~47–50% of purchases made with a discount applied.

i Hats are the most discount-sensitive product — half of all hat purchases involve a discount. This signals potential margin risk and pricing sensitivity in these categories.

Customer Segmentation

🏆 Loyal

3,116 customers — the dominant segment, representing the core repeat-purchase base.

🔄 Returning

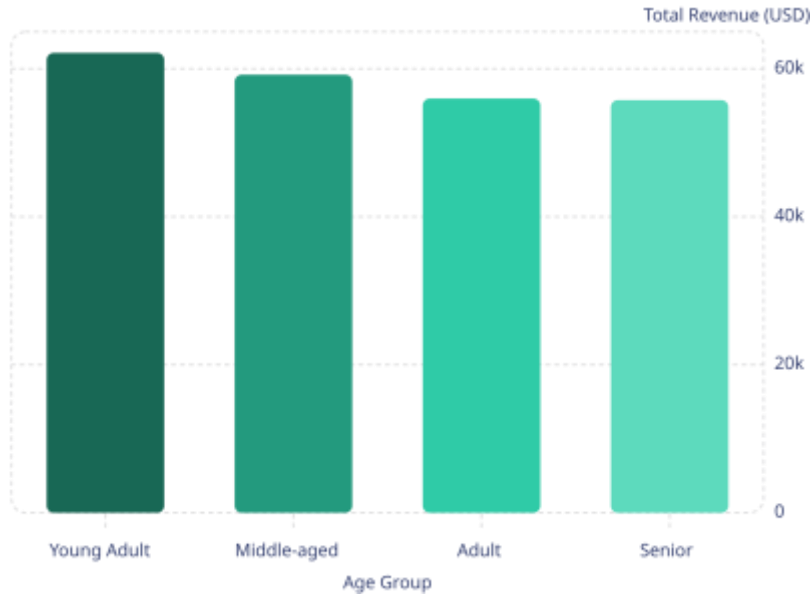
701 customers — engaged but not yet at loyal frequency. Prime targets for loyalty programs.

🆕 New

83 customers — smallest segment. Opportunity to convert through onboarding incentives.



Revenue by Age Group



Age Group Revenue Contribution

Revenue is relatively evenly distributed across age groups. **Young Adults** lead at \$62,143, while **Seniors** contribute the least at \$55,763 — a difference of only ~11%.

This suggests broad product appeal across demographics, with slight opportunity to deepen engagement with older segments.



Business Recommendations



Boost Subscriptions

Promote exclusive subscriber benefits to grow the 27% base and capture more recurring revenue.



Loyalty Programs

Reward the 701 Returning customers to accelerate conversion into the Loyal segment.



Review Discount Policy

Balance sales boosts with margin control — nearly 50% of top products rely heavily on discounts.



Targeted Marketing

Focus campaigns on high-revenue age groups, express-shipping users, and top-rated products.